Hollow 20 mins 20 mins

Directions: For questions 1-20, there are two amounts in boxes labeled "**A**" and "**B**." You will need to determine which amount, if any, is greater. If the amount in box **A** is greater, circle in choice **A** on your answer sheet. If the amount in box **B** is greater, circle in choice **B** on your answer sheet. If the amounts in boxes **A** and **B** are equal, circle in choice **C** on your answer sheet. If there is not enough information to determine which box is greater, circle in choice **D** on your answer sheet.

1. Let $a \star b = (a+b)ab$

The value of $3 \star 4$	The value of $2 \star 5$
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

2. $x + w + 3 = 21$ $\qquad$ $y + 3 + z = 19$

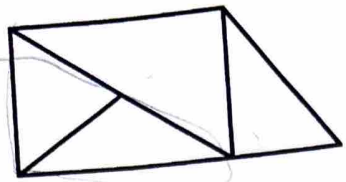
The value of w	The value of z
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

3.

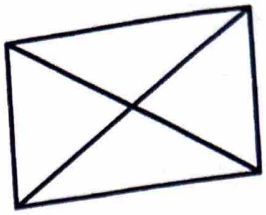
Figure A



The number of triangles in figure A.

A

Figure B



The number of triangles in figure B.

B

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

4.

8 handfuls of grain feed 4 chicks for one day.

The number of handfuls needed to feed 6 chicks for 3 days

A

The number of handfuls needed to feed 2 chicks for 7 days

B

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

5.

The largest number you can make using each of the digits 5, 7, 7, 7, and 8 one time	The largest number you can make using each of the digits 1, 2, 3, 4, 4, and 2 one time
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

6.

Figure A

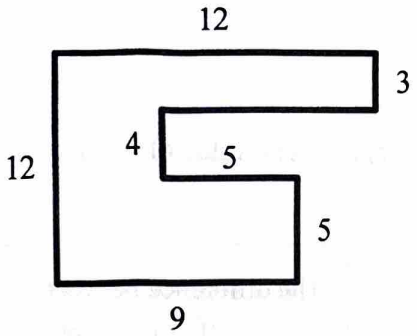
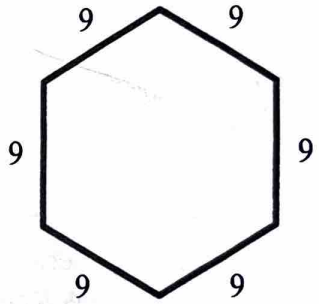


Figure B



The perimeter of figure A	The perimeter of figure B
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

7.

A	B
The number of mints left over after a pack of 28 mints is split evenly between 5 people	The number of candies left over after a pack of 26 candies is split evenly between 7 people

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

8.

Point B (not shown) lies between points A and C. The value of $B + 1$ is 16.

A	B
The difference between the values of points B and A	The difference between the values of points C and B

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

9.

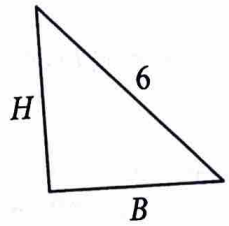
The number of zeros in 2.36×10^4	The number of zeros in 4.6×10^3
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

10. The area of the triangle below is 32 square units.

Area of a triangle = $\frac{1}{2}(B \times H)$.



(Figure not drawn to scale)

The value of the base B	The value of the height H
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

11. In the figure below, the sum of each column, row, and diagonal must be equal to 9.

4	B	4
3	3	3
2	2	A

The value of letter A	The value of letter B
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

12. Jean-Raffia goes to the grocery store once every three days. He does his laundry once every four days. Jean-Raffia went to both the grocery store and the laundromat on the first day of January. There are 31 days in the month of January.

The number of days that Jean-Raffia will go to both the grocery store and the laundromat during January	The date of the day in February when he will first go to both the grocery store and the laundromat
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

13. Derek is saving money to go on vacation. The pattern below shows the total amount he has each week.

2, 6, 18, 54 . . .

The total amount Derek has in week 5	Three times the amount Derek has in week 4
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

14. Bill is standing in line at an amusement park. There are 12 people in front of him. There are 26 people in the line.

The number of people behind him in line	Bill's place number in the line
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

Stop

15. A telephone wire is 30 feet long. It hangs between two telephone poles that are 14 feet tall.

The height of the lowest point of the wire	The distance between the two telephone poles
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

16.

	A	B	C	D	E	F
Row 1	1	2	3	4	5	6
Row 2	7	8	9	10	11	12
Row 3	13	14	15	16	17	18
Row 4	19	20	21	22	23	24
Row 5	25	26	27	28	29	30

The row number that contains 52	The number in Row 2, Column B
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

17.

The area of a rectangle that is 6 feet long and 10 feet wide	The area of a square with a perimeter equal to the perimeter of the rectangle from box **A**
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

18. The following is a sequence of digits:

2, 4, 6, 8, 2, 4, 6, 8, 2, 4, 6, 8, 2, 4 . . .

The value of the 47th digit in the sequence above	The value of the 49th digit in the sequence above
A	**B**

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

19. 2, 1, 4, 3, 6, 5, 8, 7, **A**, 9 . . .

The value of letter A

A

1, 5, 3, 7, 5, 9, 7, 11, **B**, 13 . . .

The value of letter B

B

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

20. A fish is swimming in a stream against the current. The fish swims forward 9 miles each day and drifts back 6 miles each night. Its destination is 30 miles away from where it starts.

The number of days it takes the fish to first reach a 20-mile marker upstream

A

The number of times the fish crosses the 20-mile marker on its journey

B

Which statement is true?

A. A is greater than B.
B. B is greater than A.
C. A and B are equal.
D. There is not enough information to determine which is greater.

Directions: For questions 21-35, choose the **best** answer.

21.

Each side of the scale above has 30 pounds on it. How much does each ▭ weigh if each ◊ weighs 10 pounds?

A. 5 pounds
B. 10 pounds
C. 15 pounds
D. 20 pounds

22. If 2 is subtracted from three times a number, the result is 58. What is the number?

A. 50
B. 40
C. 30
D. 20

23. Thomas is 14 years old. Two years ago, he was three years older than his brother's current age. How old was his brother five years ago?

A. 12 years old
B. 9 years old
C. 5 years old
D. 4 years old

24. In the addition problem below, what number is represented by #?

$$\begin{array}{r} 537 \\ +\ 2\#* \\ \hline 794 \end{array}$$

A. 6
B. 7
C. 5
D. 3

25. A fruit seller at the local market arranges his fruits in the following order: banana, apple, orange, strawberry. If the next fruit is a banana and he continues to arrange his fruits in the same order, what would be the 31st fruit?

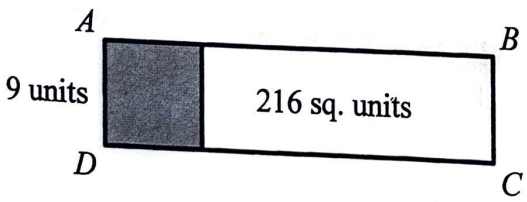
A. orange
B. apple
C. banana
D. strawberry

26. Find the sum of the digits represented by N and Z

$$\begin{array}{r} 13 \\ \times\, N \\ \hline 9Z \end{array}$$

A. 8
B. 9
C. 10
D. 11

27. In rectangle ABCD below, $\overline{AB}$ is 4 times as long as $\overline{AD}$. If the area of the white region of the rectangle is 216 square units, what is the area of the shaded region?



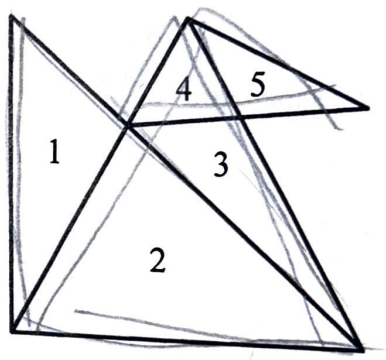
(Figure not drawn to scale)

A. 12 square units
B. 24 square units
C. 36 square units
D. 108 square units

28. Tim has an appointment to meet with Renée halfway between 12:30 PM and 1:10 AM. What time is Tim's appointment?

A. 6:50 PM
B. 12:50 PM
C. 12:50 AM
D. 9:30 PM

29. How many triangles are in the figure below?



A. 8 triangles
B. 9 triangles
C. 10 triangles
D. 11 triangles

30. What is the 50th digit in the following sequence below?

012345678901234567890123-45

A. 8
B. 9
C. 0
D. 1

31. Lena chose a random number, multiplied it by 3, divided the product by 2, and then multiplied that quotient by 4. If her final number was 96, what was her original number?

A. 12
B. 14
C. 16
D. 18

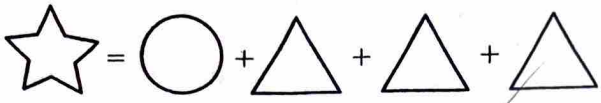
32. Cooper is a student in Mr. Talia's English class. Mr. Talia grades his tests by awarding 2 points for every correct answer and subtracting $\frac{1}{4}$ of a point for every wrong answer. How many questions did Cooper answer correctly if he answered a total of 50 questions and scored 82 points?

A. 42 questions
B. 44 questions
C. 46 questions
D. 48 questions

33. A flock of 5 birds starts to migrate south for the winter on day 1. If 3 birds join the group the next day and each day after that, how many birds will be part of the migrating flock on the 31st day of flight?

A. 90 birds
B. 93 birds
C. 95 birds
D. 115 birds

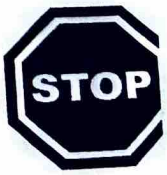
34. According to the equation below, how many triangles are there in 400 stars?



A. 400 triangles
B. 430 triangles
C. 700 triangles
D. 1,200 triangles

35. Look at the equations below.

$$5^* = 5 + 6 + 7$$
$$6^* = 6 + 7 + 8$$
$$7^* = 7 + 8 + 9$$

What is $b^* - a^*$ if $b = a + 1$ and $a = 10$?

A. 3
B. 13
C. 23
D. 10